

Math708 – Foundations of Computational Mathematics I

Fall 2011

Midterm Exam 1

You must show all of your work to get a credit for a correct answer.

- (20 points) Write down the Lagrange interpolation polynomial $p_1(x) \in \mathbb{P}_1$ for the function $f : x \rightarrow (2x - a)^4$ using the points $x_0 = 0, x_1 = a$. Show that there are two possible values for ξ in the interpolation error $f(x) - p_1(x) = \frac{\omega_2(x)f''(\xi)}{2!}$ and give their values.
- (20 points) Let $(x_0, f_0), (x_1, f_1), \dots, (x_n, f_n) \in \mathbb{R}^2$ and let $(y_0, g_0), (y_1, g_1), \dots, (y_n, g_n) \in \mathbb{R}^2$ be support points with corresponding divided differences $f[x_0, \dots, x_n]$ and $g[y_0, \dots, y_n]$. Prove that if $\{(x_k, f_k), k = 0, 1, \dots, n\} = \{(y_k, g_k), k = 0, 1, \dots, n\}$, then $f[x_0, \dots, x_n] = g[y_0, \dots, y_n]$.
- (20 points) Given the following data

x_n	0	1	2	4	6
$f(x_n)$	1	9	23	93	259

Construct a Newton's interpolation polynomial that assumes these values using divided differences. Then write the polynomial obtained above in nested form for easy evaluation (Horner's algorithm).

- (20 points) Given the following data

x_n	-1	0	$\frac{1}{2}$	1
$f(x_n)$	2	1	0	1

Construct a quadratic spline s_2 interpolating these data with additionally $s'_2(-1) = 0$.

- (30 points) Let the mapping $\mathcal{D} : \mathbb{C}^N \rightarrow \mathbb{C}^N$ defined by

$$\mathcal{D}\mathbf{c} = (-c_{k-1} + 2c_k - c_{k+1})_{k=0}^{N-1}$$

for any vector $\mathbf{c} = (c_0, c_1, \dots, c_{N-1})^T \in \mathbb{C}^N$ with $c_{-1} = c_{N-1}$ and $c_N = c_0$. Let the diagonal matrix $M = \text{diag}(\lambda_0, \lambda_1, \dots, \lambda_{N-1}) \in \mathbb{C}^{N \times N}$ where $\lambda_j = 4 \sin^2(j\pi/N) \in \mathbb{R}$ for $j = 0, 1, \dots, N-1$, and vectors $\mathbf{v}^{(j)} = (e^{i2\pi jk/N})_{k=0}^{N-1}$ for $j = 0, 1, \dots, N-1$. Then there holds

$$\mathcal{D} = \mathcal{F}^{-1} M \mathcal{F}.$$

where $\mathcal{F}$ denotes the *discrete Fourier transform*.

(Hint: Show that $\mathcal{D}\mathbf{c} = \frac{1}{N} \sum_{j=0}^{N-1} \lambda_j ((\mathbf{v}^{(j)})^H \mathbf{c}) \mathbf{v}^{(j)}$ for any $\mathbf{c} \in \mathbb{C}^N$).